

Kazi Md. Tawsif Rahman

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Academic Profile

Computer Science graduate and backend systems engineer preparing for Fall 2027 funded PhD applications. Research interests sit at the intersection of network security, privacy-preserving machine learning, trustworthy AI, and data-intensive software systems. Published work includes machine-learning-based insurance fraud detection and an IEEE conference paper on an insurance awareness mobile application with fraud detection and policy-selection features. Industry experience includes production backend systems involving REST APIs, PostgreSQL, Redis, RabbitMQ, workflow automation, rule-based assignment, audit logging, and reliable state transitions.

Research Interests

Primary: network security; privacy-preserving machine learning; trustworthy and robust AI; machine-learning-based intrusion detection systems.

Secondary: backend/distributed systems; data systems; software engineering for reliable, auditable, and scalable applications.

Research Direction for PhD Study

I am interested in building practical and reproducible security/ML systems that evaluate the tradeoff between model utility, privacy leakage, and robustness. A near-term direction is to reproduce and extend privacy-preserving ML methods for network intrusion detection using public benchmark datasets, then evaluate both predictive performance and attack-aware metrics such as membership inference risk. Long term, I want to work on deployable trustworthy AI systems where privacy, robustness, and reliability are measured together rather than treated as separate goals.

Education

Chittagong University of Engineering and Technology (CUET), Bangladesh Sep 2024
B.Sc. in Computer Science and Engineering CGPA: 3.59 / 4.00

Publications and Scholarly Work

1. **Journal Article** Kazi Md. Tawsif Rahman and Chowdhury Mahfuzul Hoq. “An Automated System for Detecting Property Insurance Fraud Using Machine Learning.” *International Journal of Mathematical Sciences and Computing (IJMSC)*, 2024. DOI: 10.5815/ijmsc.2024.03.02.
2. **IEEE Conference Paper** Mahfuzulhoq Chowdhury, Kazi Md. Tawsif Rahman, and Hossain Ahmad Maruf. “Whistle Blower: An Insurance Awareness Mobile Application with Insurance Policy Selection, Fraud Detection, Critical Help, Complaint Features.” *2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)*, pp. 1–6, 2024. IEEE. DOI: 10.1109/CSITSS64042.2024.10817002.

Research Preparation and Evidence

- **Publication experience:** contributed to two published research outputs involving applied machine learning, fraud detection, insurance technology, and software-supported decision assistance.
- **Implementation depth:** production backend experience with modular services, persistent data models, validation, caching, asynchronous processing, audit logs, and workflow-state consistency.

- **Experiment-readiness:** preparing reproducible security/ML experiments involving public NIDS datasets, baseline classifiers, differential privacy settings, and membership inference evaluation.
- **Research writing:** able to present technical problems through problem framing, method description, evaluation logic, and limitations, supported by existing publication experience.

Research Projects

Privacy-Utility Evaluation for Machine-Learning-Based Network Intrusion Detection *Planned / In Progress*

- Reproducing and analyzing privacy-preserving ML methods for network intrusion detection using public NIDS datasets such as NSL-KDD, UNSW-NB15, or CIC-IDS-2017.
- Planned evaluation includes baseline classifier performance, privacy-utility tradeoff under differential privacy, and attack-aware metrics such as membership inference risk.
- Research goal: build a reproducible benchmark connecting network security, privacy-preserving ML, and practical ML-system evaluation.
- Planned output: GitHub repository, experiment report, result tables, and professor-outreach summary.

Insurance Fraud Detection Using Machine Learning *Published Research Project*

- Developed an automated fraud-detection approach for property insurance using machine-learning methods.
- Published in *International Journal of Mathematical Sciences and Computing*; demonstrates applied ML, problem formulation, technical writing, and evaluation-oriented work.
- Research relevance: applied machine learning, fraud analytics, data-driven decision support, and trustworthy financial/insurance technology.

Whistle Blower: Insurance Awareness and Decision-Support Mobile Application *IEEE Conference Research Project*

- Co-authored IEEE CSITSS 2024 paper on a mobile application combining insurance awareness, policy selection, fraud detection, critical help, and complaint features.
- Research relevance: applied software systems, fraud detection, civic/consumer-facing decision support, and human-centered financial technology.

Reproducible Experiment Framework for Security/ML Studies *Planned*

- Planned framework for organizing datasets, model training, evaluation scripts, metrics, and experiment logs for security-oriented ML research.
- Intended to support repeatable experiments, result comparison, clean reporting, and future PhD application research projects.

Professional Experience

Software Engineer Dec 2024 – Present
BRAC IT Services Ltd., Dhaka, Bangladesh

CRM Platform

- Designed and implemented backend services for accounts, contacts, leads, opportunities, quotes, campaigns, territories, products, and pipelines, delivering 168 REST endpoints across 30 controllers.

- Built configurable rule-based assignment for leads and opportunities, including priority, schedules, conditional routing, rotation rules, duplicate checks, and validation of supported fields/operators/values.
- Improved Kanban movement scalability by replacing full-column reordering with fixed-length order keys, reducing movement updates from column-wide ranking work to single-rank assignment.
- Implemented Kanban board, column, card, metadata, and movement APIs with access control, idempotent moves, audit logs, and timeline entries.
- Added opportunity pipeline-stage Kanban support by synchronizing board columns with pipeline stages and validating allowed transitions before state changes.

Ticketing System

- Built backend services for ticket creation, updates, search, comments, forwarding, assignment rules, teams, SLA/KPI tracking, feedback, notifications, and metadata, delivering 96 REST endpoints across 21 controllers.
- Implemented least-loaded team routing using Redis-backed workload counters and a min-heap to assign tickets to resolvers with the fewest open tickets.
- Added absence-reliever handling so unavailable resolvers are skipped during assignment/reassignment and tickets fall back to relievers or other available team members.
- Designed SLA/KPI tracking through assignee-only stage updates, comment-based progress updates, status changes, notifications, and activity logs.
- Maintained resolver workload consistency across PostgreSQL and Redis through asynchronous updates after ticket creation, reassignment, and resolution.

Technical and Research Skills

Programming	Java, SQL, C, C++; add Python if actively used in research experiments
Backend / Systems	Spring Boot, Spring Security, Spring Data JPA, Hibernate, Bean Validation, Spring Modulith, Gradle, Docker, Linux
API Design	REST APIs, OpenAPI/Swagger, MapStruct, DTO design, request validation, structured error handling
Data Systems	PostgreSQL, Flyway migrations, Redis, JSONB, indexing, repository queries
Messaging / Jobs	RabbitMQ, asynchronous processing, retry handling, Quartz scheduling
Testing / Quality	JUnit 5, Mockito, API testing, Git, reproducible development workflow
Research Methods	Literature review, baseline reproduction, experiment design, result comparison, technical writing; add statistical testing/ML libraries after use
Security/ML Prep	Planned work with NIDS datasets, differential privacy, membership inference auditing, and privacy-utility evaluation

Awards and Achievements

- Codeforces Specialist; reached 1500+ rating in competitive programming.
- Finalist, CUET IUPC 2024.
- Finalist, CUET IUPC 2023.
- Finalist, CUET Tech Carnival 2020.

Selected Coursework

Algorithms and Data Structures; Database Systems; Computer Networks; Operating Systems; Software Engineering; Artificial Intelligence; Machine Learning; Information Security. *Edit this list to match your official transcript.*